The relationship between grassland species richness and the management in long-established golf courses in Japan

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OBJECTIVES

For conservation of grassland plants,to clarify the management methods and the availability oflong-established golf courses.

CONCLUSIONS

・Grassland species exist in long-established golf courses.

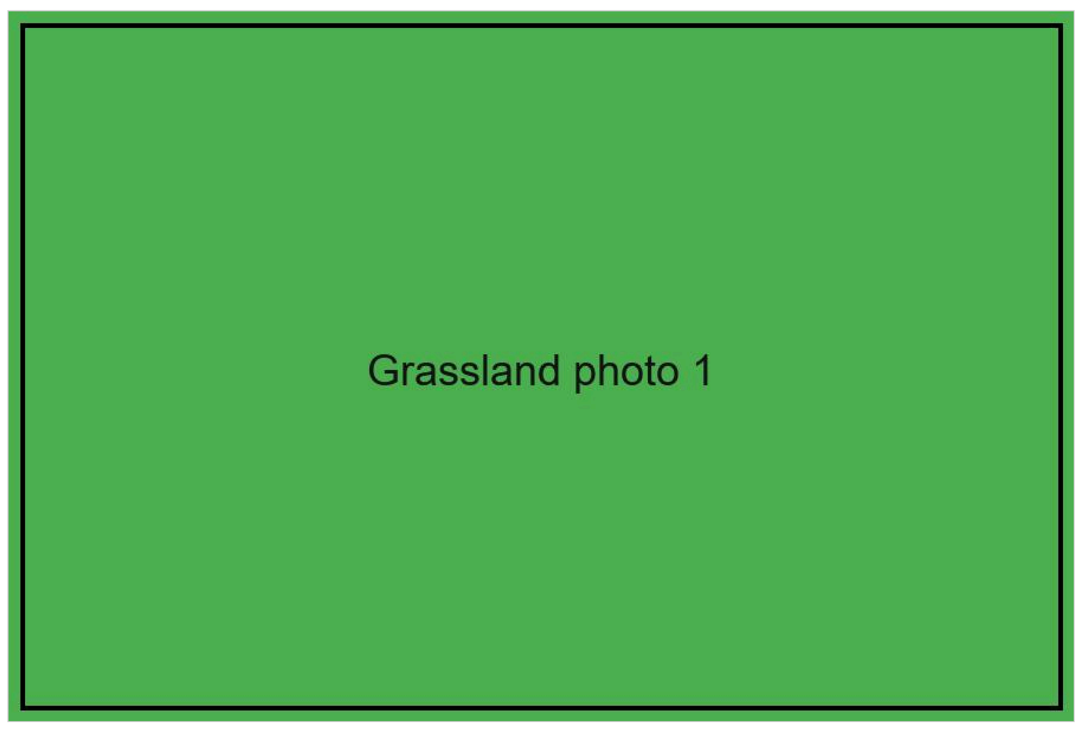
・Management of low frequency and height cutting height resulted in species rich grasslands.

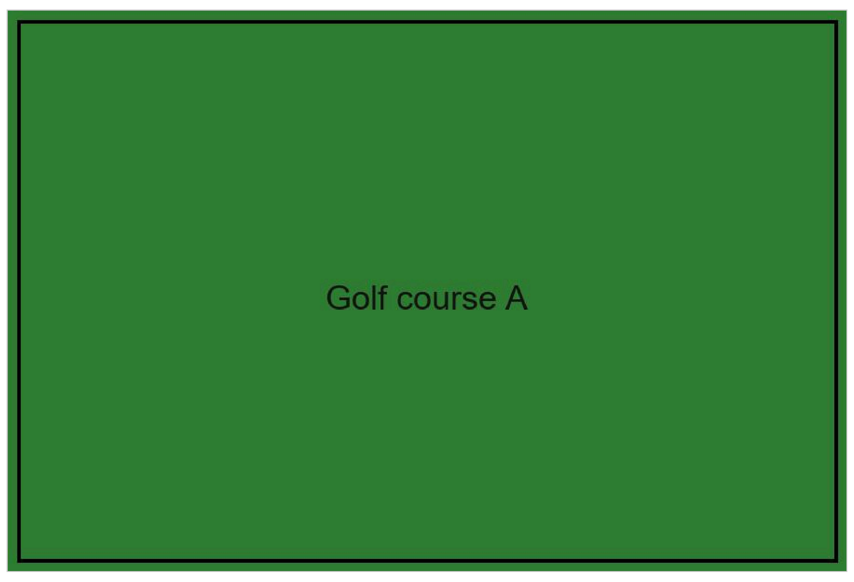
★ Golf course in appropriate management can be seed source and refugia.

BACKGROUNDS

Semi-natural grassland is one of the most important habitats.Area and species diversity of semi-natural grasslands declinedby lifestyle change and abandonment.

Remaining semi-natural grasslands are confined toagricultural margins and commercial grasslands(long-established pastures, ski slopes, golf courses).

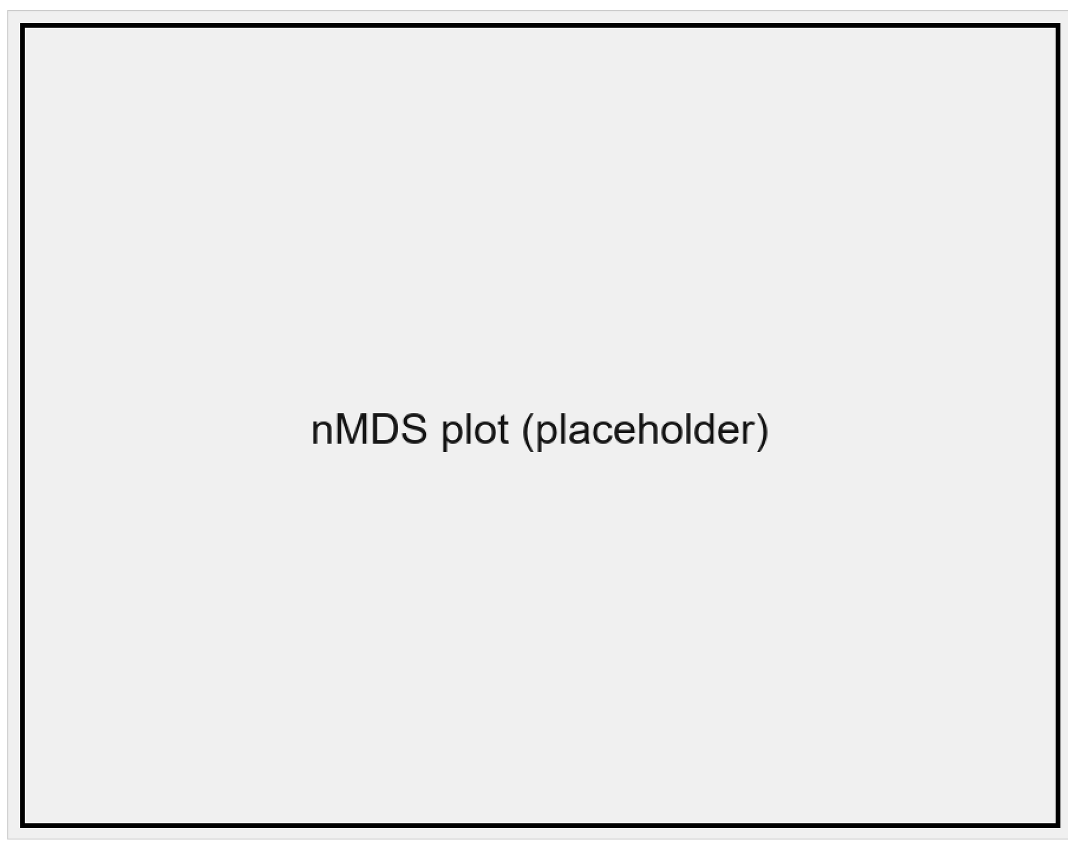




RESULTS

Species richness of each community

Community	Number of plots	Grassland Species (/m2)	All species (/m2)	Cutting height (cm)	Cutting Frequency (times/yr)
com1	57	6.1 ± 2.9	10.7 ± 4.7	12.9 ± 15.3	4.2 ± 6.1
com2	38	5.3 ± 2.7	10.6 ± 5.2	22.7 ± 19.2	4.4 ± 4.4
com3	37	3.9 ± 1.3	6.9 ± 3.1	5.1 ± 4.8	10.1 ± 5.3
com4	68	1.4 ± 0.7	2.4 ± 1.6	5.2 ± 5.0	14.6 ± 5.5



METHODS

Field investigations

・4 golf courses (established in 1903-1956)

・50 plots (1m x 1m) at rough or OB in each course

・Flora lists of grassland species in each golf course

Greenkeepers interview of management methods

・Frequency and height of mowing

・Application of fertilizer and Chemical herbicides

項目	内容
調査地	ゴルフ場4か所
方法	プロット調査 (50 plots)

SUMMARY

Summary of each golf course

Golf course	Established year	Area (ha)	Total number of grassland species	Grassland species (Mean ± SD) (/m2)	All species (Mean ± SD) (/m2)
A	1903	20	79	6.5 ± 2.8	11.2 ± 4.6
B	1926	70	55	2.9 ± 2.0	5.2 ± 4.6
C	1930	55	78	4.4 ± 2.8	8.1 ± 5.1
D	1956	60	55	2.1 ± 1.5	4.0 ± 3.3

DOMINANT SPECIES